## 1. Introduction:

The discovery of habitable planets outside our solar system has become one of the most important research areas in astronomy. With the increasing amount of astronomical data collected by space missions, it has become necessary to use advanced data analysis techniques to identify potentially habitable exoplanets.

The EXOHABITAI project focuses on applying Machine Learning techniques to analyze exoplanet datasets and predict whether a planet could be potentially habitable. By studying different planetary features such as orbital parameters, stellar properties, and environmental conditions, the system can help identify planets that may support life.

Machine learning helps in identifying hidden patterns in large datasets that are difficult to analyze manually. This project demonstrates how artificial intelligence can assist scientists in analyzing astronomical data efficiently.

## 2. Dataset Description:

The dataset used in this project contains information about different exoplanets discovered by space missions such as NASA's Kepler mission. Each entry in the dataset represents a planet candidate and includes several important parameters related to the planet and its host star.

Some common features present in the dataset include:

Orbital period of the planet

Radius of the planet

Distance from its host star

Stellar temperature

Stellar mass and luminosity

Other environmental parameters related to habitability

The dataset is stored in CSV format and contains multiple rows and columns representing different observations and features. This dataset is used as the input for training the machine learning model.

3. Data Preprocessing:

Before training the machine learning model, the dataset needs to be cleaned and prepared. Raw datasets often contain missing values, inconsistent entries, or irrelevant features that can affect the performance of the model.

The following preprocessing steps were performed:

Loading the dataset using Pandas library in Python

Handling missing values and removing incomplete records

Selecting important features required for prediction

Removing unnecessary columns that do not contribute to the model

Converting the cleaned data into a structured format suitable for machine learning algorithms

After preprocessing, the cleaned dataset was saved as a CSV file inside the data/processed folder. This processed dataset was then used for training and testing the machine learning model.

4. Machine Learning Model:

The machine learning model used in this project is the Random Forest Classifier. Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

The model training process involved the following steps:

1. Importing necessary Python libraries such as Scikit-learn, NumPy, and Pandas

2. Splitting the dataset into training data and testing data

3. Training the Random Forest model using the training dataset

4. Evaluating the model using the testing dataset

5. Generating predictions based on input features

Random Forest works by building several decision trees during the training phase. Each tree makes its own prediction, and the final result is determined by combining the predictions from all trees. This approach improves the reliability and accuracy of the predictions.

The trained model was saved as a pickle file (random_forest.pkl) so that it can be reused later without retraining the model.

5. Model Output and Results:

After training the machine learning model, predictions were generated using the testing dataset. The output results represent the predicted classification of whether a planet may be habitable or not based on the given features.

The prediction results were stored in a CSV file inside the data/processed folder. This output file contains both the actual values and predicted values produced by the model.

The successful generation of prediction results indicates that the machine learning model is functioning correctly and can analyze exoplanet data effectively.

## 6. Tools and Technologies Used

The following tools and technologies were used for implementing the EXOHABITAI project:

Python – Programming language used for model development

Pandas – Used for data manipulation and preprocessing

NumPy – Used for numerical operations

Scikit-learn – Used for implementing the Random Forest algorithm

VS Code / Jupyter Notebook – Used for writing and executing the code

CSV files – Used for storing datasets and prediction outputs

These tools helped in efficiently processing the data and building the machine learning model.

## 7. Conclusion

The EXOHABITAI project demonstrates how machine learning can be applied in the field of astronomy to analyze exoplanet datasets and

identify potentially habitable planets. By using the Random Forest algorithm, the system can analyze multiple planetary features and generate predictions based on the trained model.

Machine learning significantly improves the efficiency of analyzing large astronomical datasets. This project highlights the importance of artificial intelligence in modern space research and opens possibilities for further exploration in the search for habitable planets beyond our solar system.